Douglas Fritz

MD/PhD Researcher | Climate, Health & Infectious Disease

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Curriculum Vitae
October 2026

I work at the intersection of **climate change, infectious disease, medicine, and health-professional education**. As an NIH Oxford–Cambridge Scholar, my doctoral research combines experimental virology, epidemiology, and evolutionary modelling. I develop phylodynamic methods to estimate differences in fitness among co-circulating dengue lineages and, in a separate strand of the project, test how temperature may affect viral stability and fitness. As National Chair of Medical Students for a Sustainable Future, I help translate climate-health evidence into education, communication, and action across health-professional networks.

Climate & Health Leadership

2024 **Conference Planning Committee**
MS4SF Equity Day & MSCCH Annual Meeting

2022-Present **National Chair**
Medical Students for a Sustainable Future

2022-Present **Vice President, CU CHASE Climate Student Group**
University of Colorado

2021-Present **CU Student Sustainability Representative**
University of Colorado

2018-2019 **Sustainability Committee Creator and Co-Chair**
Saint Louis University

2017-2019 **Assistant Sustainability Coordinator**
Saint Louis University

Selected Climate & Health Initiatives

Denver Climate Risk Inventory
City and County of Denver
- ➤ Developed a risk-assessment tool to communicate neighbourhood-level climate-related health stress in Denver.

Climate Health Integration Project
CU Anschutz
- ➤ Led work to integrate planetary-health education across all four years of the medical curriculum.

Climate Grand Rounds
CU Anschutz
- ➤ Developed a climate and health grand-rounds programme for an academic medical community.

MS4SF research mentorship pipeline
Medical Students 4 a Sustainable Future
- ➤ Built connections between investigators and students pursuing climate-change and health research.

Education & Research Focus

2028 **PhD** Cambridge, UK / Bethesda, MD
University of Cambridge/NIH
- ➤ Via NIH Oxford–Cambridge Scholars Programme
- ➤ Project: Temperature Impacts on Dengue Virus Stability and Evolution
- ➤ Mentors: Leah Katzelnick (NIAID), Patrick Dolan (NIAID), Henrik Salje (UK)

2029 **MD/PhD** Denver, CO
University of Colorado Anschutz School of Medicine
- ➤ Lab Rotation: AI Computational Modeling of Single Cell Data: Fan Zhang Ph.D., Center for AI
- ➤ Lab Rotation: Viral Evolution HIV Studies: Sara Sawyer Ph.D., Biofrontiers Institute

2019 **BS Biology** St Louis, MO
Saint Louis University
- ➤ Biology Capstone Evaluating Adjuvants for a Th17-directed Chagas Vaccine
- ➤ Capstone Advisors: David Hoft, Ph.D., Lindzy Dodson, Ph.D

2019 **BS American Studies** St. Louis, MO
Saint Louis University
- ➤ American Studies Capstone: Not Just a Mom: Gender and Maternity in the West Lake Landfill Crisis
- ➤ Capstone Advisors: Kate Moran, Ph.D.

Selected Climate, Health & Infectious-Disease Scholarship

Chen, Y., **Fritz, D.**, Clapham, H., Lefrancq, N., Salje, H., and the BUZZ study team (2026). Uncovering the fitness of endemically circulating Zika virus strains [In Review] *Nature Communications*

Mayer, S. V., Winkelmann, E. R., Coello Escoto, A., Dorman, J., McCracken, M., **Fritz, D.**, Baric, R., Tse, V., Cummings, D., Whitehead, S., Best, S., Dolan, P., Aaskov, J., Gromowski, G., Katzelnick, L. C. (2026). Evolutionarily distinct dengue virus 2 with a unique type-specific neutralization determinant [In Review] *PLOS Pathogens*

Perez Duque, M., Lefrancq, N., **Fritz, D.**, Salje, H. (2026). Locally structured transmission and lineage-level turnover underlie Japanese encephalitis virus ecology [In Review] *Nature Medicine*

Smith, J., Canty, W. K., Blevins, K., Walsh, C. J., **Fritz, D.**, Foy, B., Olson, N., Shelby, E. R., Gillespie, E. J. (2026). Integrating climate medicine into US medical education through a five-step planetary health curriculum development process *Discover Education* 5. https://doi.org/10.1007/s44217-026-01782-6

Potarazua, S., **Fritz, D.**, Drew, T., Malani, K., Randle, H., Bhardwaj, N., McKernan, K., Melnyk, A. (2026). At the Crossroads of Climate Change and Reproductive Choice: A Scoping Review [In Review] *Journal of Climate Change and Health*

Azar M Abadi, Kyle D. Brumfield, Moiz Usmani, … Fritz, D, et al. (2026). Developing Scenario-Based Strategies for Health, Climate, and Environmental Preparedness: The One Health, One Earth Approach *GeoHealth* 10. https://doi.org/10.1029/2025GH001674

Marwah, H., Hampshire, K., Sood, N., **Fritz, D.**, Brewer, T., Walden, D., Potarazu, S.. (2023). Student-driven climate action to safeguard health *eBioMedicine* 93. https://doi.org/10.1016/j.ebiom.2023.104693

Fritz, D., Nuzhat, I. (2022). "OSS15.4 Arboviruses and Climate Change." *Climate Resources for Health Education* https://climatehealthed.org/course/arboviruses/

Selected Climate & Health Presentations

Fritz, D. (2026, January). *Extreme Heat & Health: Making an Action Plan* [Invited oral presentation]. School of Medicine University of Glasgow.

Fritz, D., Katzelnick, L. (2025, February). *Effect of Temperature on dengue dynamics: NIH New Proposal GPP 2025* [*1st Place Poster & 1st Place 3 Min. Talk* Oral Presentation]. NIH Graduate Student Research Symposium.

Fritz, D., Walsh, C., Blevins, K., Canty, L., Foy, B., Smith, J., Gillespie, E. (2024, April). *Planetary Health Curriculum Integration: A Student-led, Results Driven Pipeline to LO-Based Integration* [Poster presentation]. AAMC 2024 Group on Student Affairs (GSA), Careers in Medicine (CiM), Organization of Student Representatives (OSR) National Meeting.

Blevins, K., Walsh, C., Smith, J., Canty, L., Foy, B., **Fritz, D.**, Gillespie, E. MD (2024, March). *Developing Physicians of the Future: Interactive Planetary Health Based Medical School Course Spurs Confidence In Climate Sensitive Healthcare* [Poster presentation]. CU Anschutz C4Health Conference.

Fritz, D. (2024, January). *Why and How You (as a Med Student) Should be Talking to Your Patients about Climate Change and Their Health* [Oral Presentation]. Medical Student Grand Rounds - CU Anschutz.

Smith, J., Blevins, K., Canty, L., Foy, B., **Fritz, D.**, Walsh, C., Gillespie, E. MD (2024, February). *Planetary Health Curriculum Integration: A student-led stepwise process to teach climate medicine at your Med School.* [Poster presentation]. MS4SF's Climate Health Equity Day.

Fritz, D. (2023, September). *Keynote: Why and How you should be talking to your patients about climate change* [Keynote]. AHEC NEAR Conference.

Fritz, D., Blevins, K., Smith, J., Canty, L., Walsh, C., Gillespie, E. (2023, September). *Advances in Planetary Health Curriculum Integration.* [Poster presentation]. AHEC NEAR Conference.

Potarazu, S., Brewer, T., **Fritz, D.**, Marwah, H., Sood, N., Diedrich, T., Malits, J., Manivannan, M. (2023, April). *AAMC Webinars and Online Courses: Exploring Student Perspectives: The Current Landscape of Student- led Initiatives in Climate & Health Education (Building Better Curriculum)* [Webinar]. AAMC Webinars and Online Courses:.

Potarazu, S., **Fritz, D.**, Malani, K., Drew, T., Randle, H., Bhardwaj, N., McKernan, K., Melnyk, A. (2023, October). *At the Crossroads of Climate Change and Reproductive Choice: A Scoping Review* [Conference session]. Climate and Health 2023.

Research & Professional Experience

2021-2022 **Epidemiologist 1**
Kantar Health/Cerner-Envisa

2020-2021 **IRTA Postbaccalaureate Fellow**
Ted Pierson Lab, National Institute of Allergy and Infectious Disease

2019-2020 **Fulbright Scholar**
Goce Delchev University (UGD)

2016-2019 **Infectious Disease Scientist**
Daniel Hoft Lab, Saint Louis University Medical School

2016-2017 **Microflora Research Assistant**
Biology Department, Saint Louis University

2015 **Research Assistant**
Monsanto

Selected Awards, Fellowships & Grants

- NIH Graduate Student Poster Day, 1st Place (2025)
- NIH Graduate Student Three-Minute Talk Competition, 1st Place (2025)
- Physicians for Social Responsibility Next Generation Climate & Health Ambassador (2022)
- Fulbright Scholar, North Macedonia (2019–2020)

2022 **Interest Group Outreach Grant: Climate Related Stress Assessment in Denver CO**
Colorado Family Medicine Association
- $1,500

2024 **Fulbright Colloquium**
Fulbright
- $11,500

Relevant Methods

Bioinformatics: scRNA-seq, spatial transcriptomics
Code Languages: R, SQL, Python, HTML (Basic), Excel Macro
Data Visualization: R, Shiny, Prism, Tableau, Illustrator
Statistical Skills: Real-World Experience: Gaussian and non-gaussian regression modeling, multilevel modelling„ machine learning, survey design
Wet Lab: Enzyme linked assays, immunoassay binding assays, Flow Cytometry/CellSort, SDS-Page, gel blots, HPLC, GC-MS, cell cultures, qRTPCR